Interactive Constraint Learning from Data with Hybrid AI

- **Keywords** : Hybrid AI, machine learning, constraint programming, pattern mining, human-in-the-loop learning
- **Location:** IMT Atlantique, Nantes Campus, France
- **Research team:** TASC, LS2N Laboratory
- **Funding:** ANR FIDD project
- **Duration:** 36 months
- **Application deadline:** August 30, 2026
- **Expected starting date:** Autumn 2026, with some flexibility
- **Supervision** :
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PhD project

How can an AI system automatically learn the constraints, preferences, and decision rules that matter to a user?

Constraint programming provides powerful and expressive tools for modelling complex data-mining and decision-support problems. However, defining the relevant constraints still requires significant expertise and considerable interaction with domain specialists.

This PhD aims to develop **hybrid AI methods combining machine learning, pattern mining, and constraint programming** to learn interpretable symbolic constraints directly from data and user feedback. The objective is to reduce the effort required to build constraint-based models while ensuring that the learned knowledge remains understandable, editable, and reusable by domain experts.

The PhD is part of the ANR-funded **FIDD project**, which investigates interactive and constraint-driven methods for facilitated data exploration.

Research challenges

The research will address four main challenges.

1. **Discover informative representations from data**. The first objective will be to identify concise and representative patterns from positive and negative examples. These patterns will provide informative features for learning the constraints that best capture the user's expectations.
2. **Learn interpretable symbolic constraints.** The PhD will investigate supervised and weakly supervised approaches for learning expressive Boolean formulas involving:
   - numerical constraints with unknown thresholds;
   - arithmetic relations between variables;
   - set-based conditions requiring the discovery of relevant item-sets;
   - conjunctions and disjunctions of parameterized constraints.
3. **Learn interactively from user feedback**. The proposed methods will progressively refine the learned constraints through interaction with the user. A central objective will be to select the most informative questions or examples while limiting the amount of feedback required from the expert. This human-in-the-loop process should allow users to validate, reject, or adjust constraints and their parameters.
4. **Handle imperfect and evolving data**. Real-world data may contain noisy measurements, missing values, or incorrectly labeled examples. The PhD will therefore study robust methods for learning constraints that provide a meaningful trade-off between predictive accuracy, interpretability, robustness to noise and number of user interactions.

Real-world application

The methods will be evaluated on a real-world use case developed in collaboration with **IMADIS Group**, in the field of emergency teleradiology.

The objective will be to learn interpretable constraints, preferences, and decision rules from operational data and expert feedback. This application will allow the successful candidate to address fundamental research questions while working on a concrete problem involving healthcare workflows and decision support.

Research environment

The successful candidate will join the TASC research team within the LS2N laboratory at IMT Atlantique's Nantes campus.

The TASC team conducts research in constraint programming, data mining, decision support, hybrid symbolic AI, and Human–AI interaction. The team also contributes to the development of Choco Solver, an internationally recognized open-source constraint-programming library.

The PhD student will collaborate with researchers from IMT Atlantique, LISN, and the academic and industrial partners of the FIDD project. The project will offer opportunities to:

- develop theoretical and algorithmic contributions;
- implement open-source research software;
- experiment with real-world datasets;
- present results at leading international conferences in artificial intelligence, machine learning, constraint programming, and data mining.

Candidate profile

Applicants should have completed, or be in the final stage of completing, a Master's degree in computer science, artificial intelligence, applied mathematics, or a related field. We are looking for a candidate with:

- a strong interest in research and scientific problem solving;
- solid programming and algorithmic skills;
- an interest in machine learning, data mining, constraint programming, or symbolic AI;
- good written and oral communication skills in English.

Experience in one or more of the following areas would be appreciated, but is not mandatory:

- constraint programming or combinatorial optimization;
- machine learning or symbolic learning;
- pattern mining and knowledge discovery;
- Java or Python;
- constraint solvers such as Choco Solver, OR-Tools, or related frameworks.

Candidates are not expected to already be experts in all these areas. The PhD is particularly suitable for applicants interested in developing skills at the intersection of symbolic and data-driven AI.

Application procedure

Applications should include:

- a cover letter describing the candidate's motivation, research interests, and suitability for the position;
- an up-to-date CV;
- academic transcripts;
- the contact details of one or two academic referees.

Applications and requests for further information should be sent to the three supervisors.

Selected references

1. Bessiere et al. *Learning Constraints through Partial Queries*. Artificial Intelligence, 2023.

2. Guns, Nijssen, and De Raedt. *Itemset Mining: A Constraint Programming Perspective*. Artificial Intelligence, 2011.

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4. Lazaar et al. *A Global Constraint for Closed Frequent Pattern Mining*. CP, 2016.

5. Prud'homme and Fages. *Choco-Solver: A Java Library for Constraint Programming*. JOSS, 2022.

6. Vernerey et al. *Threshold-Free Pattern Mining Meets Multi-objective Optimization*. IJCAI, 2022.